

БДЗ по дискретной математике №2

Б18-504

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Задание 1

- $[u/x, v/y](\forall x(\exists y P(x, y) \vee Q(y, x)) \supset (\exists z P(x, y) \vee \forall x Q(y, x)))$

Выполним подстановку:

1. $[u/x, v/y]\forall x(\exists y P(x, y) \vee Q(y, x)) \supset [u/x, v/y](\exists z P(x, y) \vee \forall x Q(y, x))$
2. $\forall x[v/y](\exists y P(x, y) \vee Q(y, x)) \supset ([u/x, v/y]\exists z P(x, y) \vee [u/x, v/y]\forall x Q(y, x))$
3. $\forall x([v/y]\exists y P(x, y) \vee [v/y]Q(y, x)) \supset (\exists z[u/x, v/y]P(x, y) \vee \forall x[v/y]Q(y, x))$
4. $\forall x(\exists y P(x, y) \vee Q(v, x)) \supset (\exists z P(u, v) \vee \forall x Q(v, x))$

- $[z/x, z/y](\forall x(\exists y P(x, y) \vee Q(y, x)) \supset (\exists z P(x, y) \vee \forall x Q(y, x)))$

Выполним подстановку:

1. $[z/x, z/y]\forall x(\exists y P(x, y) \vee Q(y, x)) \supset [z/x, z/y](\exists z P(x, y) \vee \forall x Q(y, x))$
2. $\forall x[z/y](\exists y P(x, y) \vee Q(y, x)) \supset ([z/x, z/y]\exists z P(x, y) \vee [z/x, z/y]\forall x Q(y, x))$
3. $\forall x([z/y]\exists y P(x, y) \vee [z/y]Q(y, x)) \supset (\exists z'[z/x, z/y]P(x, y) \vee \forall x[z/y]Q(y, x))$
4. $\forall x(\exists y P(x, y) \vee Q(z, x)) \supset (\exists z' P(z, z) \vee \forall x Q(z, x))$

Задание 2

- $[R(x,y) \supset R(y,y)]_{P(x,y)}(P(x, y) \wedge Q(z, y) \supset P(y, x) \vee R(z, y))$

Выполним подстановку:

1. $[R(y,x) \supset R(y,y)]_{P(x,y)}(P(x, y) \wedge Q(z, y)) \supset [R(y,x) \supset R(y,y)]_{P(x,y)}(P(y, x) \vee R(z, y))$
2. $[R(y,x) \supset R(y,y)]_{P(x,y)}P(x, y) \wedge [R(y,x) \supset R(y,y)]_{P(x,y)}Q(z, y) \supset [R(y,x) \supset R(y,y)]_{P(x,y)}P(y, x) \vee [R(y,x) \supset R(y,y)]_{P(x,y)}R(z, y)$
3. $(R(y, x) \supset R(y, y)) \wedge Q(z, y) \supset (R(x, y) \supset R(x, x)) \vee R(z, y)$

- $[R(x,y) \supset R(z,y)]_{P(x,y)}(\neg P(z, z) \supset Q(x, x) \supset \forall z(P(y, x) \supset Q(z, y)))$

Выполним подстановку:

1. $[R(x,y) \supset R(z,y)]_{P(x,y)}\neg P(z, z) \supset [R(x,y) \supset R(z,y)]_{P(x,y)}Q(x, x) \supset [R(x,y) \supset R(z,y)]_{P(x,y)}\forall z(P(y, x) \supset Q(z, y))$
2. $\neg[R(x,y) \supset R(z,y)]_{P(x,y)}P(z, z) \supset Q(x, x) \supset \forall z'[R(x,y) \supset R(z,y)]_{P(x,y)}(P(y, x) \supset Q(z, y))$
3. $\neg(R(z, z) \supset R(z, z)) \supset Q(x, x) \supset \forall z'([R(x,y) \supset R(z,y)]_{P(x,y)}P(y, x) \supset [R(x,y) \supset R(z,y)]_{P(x,y)}Q(z', y))$
4. $\neg(R(z, z) \supset R(z, z)) \supset Q(x, x) \supset \forall z'(R(y, x) \supset R(z, x) \supset Q(z', y))$

- $[\forall z(R(x,y) \supset R(z,y)) / P(x,y)](\neg P(z,z) \supset Q(x,x) \supset \forall z(P(y,x) \supset Q(z,y)))$
 1. $[\forall z(R(x,y) \supset R(z,y)) / P(x,y)] \neg P(z,z) \supset [\forall z(R(x,y) \supset R(z,y)) / P(x,y)] Q(x,x) \supset$
 $[\forall z(R(x,y) \supset R(z,y)) / P(x,y)] \forall z(P(y,x) \supset Q(z,y))$
 2. $\neg [\forall z(R(x,y) \supset R(z,y)) / P(x,y)] P(z,z) \supset Q(x,x) \supset$
 $\forall z [\forall z(R(x,y) \supset R(z,y)) / P(x,y)] (P(y,x) \supset Q(z,y))$
 3. $\neg [z/x, z/y] \forall z(R(x,y) \supset R(z,y)) \supset Q(x,x) \supset$
 $\forall z ([\forall z(R(x,y) \supset R(z,y)) / P(x,y)] P(y,x) \supset [\forall z(R(x,y) \supset R(z,y)) / P(x,y)] Q(z,y))$
 4. $\neg \forall z' [z/x, z/y] (R(x,y) \supset R(z',y)) \supset Q(x,x) \supset$
 $\forall z ([y/x, x/y] \forall z(R(x,y) \supset R(z,y)) \supset Q(z,y))$
 5. $\neg \forall z' ([z/x, z/y] R(x,y) \supset [z/x, z/y] R(z',y)) \supset Q(x,x) \supset$
 $\forall z (\forall z [y/x, x/y] (R(x,y) \supset R(z,y)) \supset Q(z,y))$
 6. $\neg \forall z' (R(z,z) \supset R(z',z)) \supset Q(x,x) \supset$
 $\forall z (\forall z ([y/x, x/y] R(x,y) \supset [y/x, x/y] R(z,y)) \supset Q(z,y))$
 7. $\neg \forall z' (R(z,z) \supset R(z',z)) \supset Q(x,x) \supset \forall z (\forall z (R(y,x) \supset R(z,x)) \supset Q(z,y))$

Задание 3 $\forall x(A \equiv B) \supset (\forall x(B \equiv C) \supset \forall x(A \equiv C))$

Доказательство с использованием свойств секвенций.

1. $\vdash \forall x(A \equiv B) \supset (\forall x(B \equiv C) \supset \forall x(A \equiv C))$ $\supset_R$
2. $\forall x(A \equiv B) \vdash \forall x(B \equiv C) \supset \forall x(A \equiv C)$ $\supset_R$
3. $\forall x(A \equiv B), \forall x(B \equiv C) \vdash \forall x(A \equiv C)$ $\forall_R$
4. $\forall x(A \equiv B), \forall x(B \equiv C) \vdash [c_1/x]A \equiv [c_1/x]C$ $\forall_{L,L}$
5. $\forall x(A \equiv B), \forall x(B \equiv C), [c_1/x]A \equiv [c_1/x]B, [c_1/x]B \equiv [c_1/x]C \vdash [c_1/x]A \equiv [c_1/x]C$ $W_{L,L}$
6. $[c_1/x]A \equiv [c_1/x]B, [c_1/x]B \equiv [c_1/x]C \vdash [c_1/x]A \equiv [c_1/x]C$ $Def_{\equiv}$
7. $[c_1/x]A \equiv [c_1/x]B, [c_1/x]B \equiv [c_1/x]C \vdash ([c_1/x]A \supset [c_1/x]C) \wedge ([c_1/x]C \supset [c_1/x]A)$ $\wedge_R$
8. (*) | 9. (**)

(*)	
8. $[c_1/x]A \equiv [c_1/x]B, [c_1/x]B \equiv [c_1/x]C \vdash [c_1/x]A \supset [c_1/x]C$	$Def_{\equiv}, \wedge_L$
10. $[c_1/x]A \supset [c_1/x]B, [c_1/x]B \supset [c_1/x]A, [c_1/x]B \equiv [c_1/x]C \vdash [c_1/x]A \supset [c_1/x]C$	W_L
11. $[c_1/x]A \supset [c_1/x]B, [c_1/x]B \equiv [c_1/x]C \vdash [c_1/x]A \supset [c_1/x]C$	$Def_{\equiv}, \wedge_L$
12. $[c_1/x]A \supset [c_1/x]B, [c_1/x]B \supset [c_1/x]C, [c_1/x]C \supset [c_1/x]B \vdash [c_1/x]A \supset [c_1/x]C$	W_L
13. $[c_1/x]A \supset [c_1/x]B, [c_1/x]B \supset [c_1/x]C \vdash [c_1/x]A \supset [c_1/x]C$	$\supset_L$
15.	
$[c_1/x]B, [c_1/x]B \supset [c_1/x]C \vdash [c_1/x]A \supset [c_1/x]C$	
$\supset_R$	
17.	
$[c_1/x]B, [c_1/x]B \supset [c_1/x]C, [c_1/x]A \vdash [c_1/x]C$	
W_L	
18.	
$[c_1/x]B, [c_1/x]B \supset [c_1/x]C \vdash [c_1/x]C$	
$\supset_L$	
14.	
$[c_1/x]B \supset [c_1/x]C \vdash [c_1/x]A, [c_1/x]A \supset [c_1/x]C$	
$\supset_R$	
16.	
$[c_1/x]B \supset [c_1/x]C, [c_1/x]A \vdash [c_1/x]A, [c_1/x]C$	
$W_{L,R}$	
17.	
$[c_1/x]A \vdash [c_1/x]A$	
(акс.)	
19.	
$[c_1/x]B \vdash [c_1/x]C, [c_1/x]B$	
W_R	
21.	
$[c_1/x]B \vdash [c_1/x]B$	
(акс.)	
20.	
$[c_1/x]B, [c_1/x]C \vdash [c_1/x]C$	
W_L	
22.	
$[c_1/x]C \vdash [c_1/x]C$	
(акс.)	

(**)		
9.	$[c_1/x]A \equiv [c_1/x]B, [c_1/x]B \equiv [c_1/x]C \vdash [c_1/x]C \supset [c_1/x]A$	$Def_{\equiv}, \wedge_L$
23.	$[c_1/x]A \supset [c_1/x]B, [c_1/x]B \supset [c_1/x]A, [c_1/x]B \equiv [c_1/x]C \vdash [c_1/x]C \supset [c_1/x]A$	W_L
24.	$[c_1/x]B \supset [c_1/x]A, [c_1/x]B \equiv [c_1/x]C \vdash [c_1/x]C \supset [c_1/x]A$	$Def_{\equiv}, \wedge_L$
25.	$[c_1/x]B \supset [c_1/x]A, [c_1/x]B \supset [c_1/x]C, [c_1/x]C \supset [c_1/x]B \vdash [c_1/x]C \supset [c_1/x]A$	W_L
26.	$[c_1/x]B \supset [c_1/x]A, [c_1/x]C \supset [c_1/x]B \vdash [c_1/x]C \supset [c_1/x]A$	$\supset_L$
27.	$[c_1/x]C \supset [c_1/x]B \vdash [c_1/x]C \supset [c_1/x]A, [c_1/x]B$	
	$\supset_R$	
29.	$[c_1/x]C \supset [c_1/x]B, [c_1/x]C \vdash [c_1/x]A, [c_1/x]B$	
	W_R	
30.	$[c_1/x]C \supset [c_1/x]B, [c_1/x]C \vdash [c_1/x]B$	
	$\supset_L$	
31.	$[c_1/x]C \vdash [c_1/x]B, [c_1/x]C$	
	W_R	
33.	$[c_1/x]C \vdash [c_1/x]C$	
	(акс.)	
32.	$[c_1/x]B, [c_1/x]C \vdash [c_1/x]B$	
	W_L	
34.	$[c_1/x]B \vdash [c_1/x]B$	
	(акс.)	
28.	$[c_1/x]A, [c_1/x]C \supset [c_1/x]B \vdash [c_1/x]C \supset [c_1/x]A$	
	$\supset_R$	
35.	$[c_1/x]A, [c_1/x]C \supset [c_1/x]B, [c_1/x]C \vdash [c_1/x]A$	
	$W_{L,R}$	
36.	$[c_1/x]A \vdash [c_1/x]A$	
	(акс.)	

Задание 4 $\exists x(A \supset B) \supset (\forall xA \supset \exists xB)$

Доказательство с использованием свойств секвенций.

1.	$\vdash \exists x(A \supset B) \supset (\forall xA \supset \exists xB)$	$\supset_R$
2.	$\exists x(A \supset B) \vdash \forall xA \supset \exists xB$	$\supset_R$
3.	$\exists x(A \supset B), \forall xA \vdash \exists xB$	$\exists_L$
4.	$[c_1/x]A \supset [c_1/x]B, \forall xA \vdash \exists xB$	$\supset_L$
5.	$[c_1/x]B, \forall xA \vdash \exists xB$	W_L
7.	$[c_1/x]B \vdash \exists xB$	$\exists_L$
8.	$[c_1/x]B \vdash [c_1/x]B, \exists xB$	W_R
9.	$[c_1/x]B \vdash [c_1/x]B$	(акс.)
6.	$\forall xA \vdash \exists xB, [c_1/x]A$	W_R
10.	$\forall xA \vdash [c_1/x]A$	$\forall_L$
11.	$\forall xA, [c_1/x]A \vdash [c_1/x]A$	W_L
12.	$[c_1/x]A \vdash [c_1/x]A$	(акс.)

Задание 5

- $\forall x(A \wedge B) \equiv \forall xA \wedge \forall xB$

Доказательство.

1.	$\vdash (\forall x(A \wedge B) \supset \forall xA \wedge \forall xB) \wedge (\forall xA \wedge \forall xB \supset \forall x(A \wedge B))$	$\wedge_R$
2.	$\vdash \forall x(A \wedge B) \supset \forall xA \wedge \forall xB$	$\supset_R$
4.	$\forall x(A \wedge B) \vdash \forall xA \wedge \forall xB$	$\wedge_R$
5.	(*)	
6.	(**)	
3.	$\vdash \forall xA \wedge \forall xB \supset \forall x(A \wedge B)$	$\supset_R$
15.	$\forall xA \wedge \forall xB \vdash \forall x(A \wedge B)$	$\wedge_R$
16.	$\forall xA, \forall xB \vdash \forall x(A \wedge B)$	$\forall_R$
17.	$\forall xA, \forall xB \vdash [c_3/x]A \wedge [c_3/x]B$	$\wedge_R$
18.	(***)	
19.	(****)	

5. $\forall x(A \wedge B) \vdash \forall x A$ $\forall_R$
7. $\forall x(A \wedge B) \vdash [c^1/x]A$ $\forall_L$
- (*) 8. $\forall x(A \wedge B), [c^1/x]A \wedge [c^1/x]B \vdash [c^1/x]A$ $\wedge_L$
9. $\forall x(A \wedge B), [c^1/x]A, [c^1/x]B \vdash [c^1/x]A$ $W_{L,L}$
10. $[c^1/x]A \vdash [c^1/x]A$ (акс.)

6. $\forall x(A \wedge B) \vdash \forall x B$ $\forall_R$
11. $\forall x(A \wedge B) \vdash [c^2/x]B$ $\forall_L$
- (**) 12. $\forall x(A \wedge B), [c^2/x]A \wedge [c^2/x]B \vdash [c^2/x]B$ $\wedge_L$
13. $\forall x(A \wedge B), [c^2/x]A, [c^2/x]B \vdash [c^2/x]B$ $W_{L,L}$
14. $[c^2/x]B \vdash [c^2/x]B$ (акс.)

18. $\forall x A, \forall x B \vdash [c^3/x]A$ $\forall_L$
- (***) 20. $\forall x A, \forall x B, [c^3/x]A \vdash [c^3/x]A$ $W_{L,L}$
21. $[c^3/x]A \vdash [c^3/x]A$ (акс.)

19. $\forall x A, \forall x B \vdash [c^3/x]B$ $\forall_L$
- (****) 22. $\forall x A, \forall x B, [c^3/x]A \vdash [c^3/x]B$ $W_{L,L}$
23. $[c^3/x]B \vdash [c^3/x]B$ (акс.)

- $\exists x(A \vee B) \equiv \exists x A \vee \exists x B$

Доказательство.

1. $\vdash (\exists x(A \vee B) \supset \exists x A \vee \exists x B) \wedge (\exists x A \vee \exists x B \supset \exists x(A \vee B)) \quad \wedge_R$
2. (*) | 3. (**)

- (*)
2. $\vdash \exists x(A \vee B) \supset \exists x A \vee \exists x B \quad \supset_R$
4. $\exists x(A \vee B) \vdash \exists x A \vee \exists x B \quad \forall_R$
5. $\exists x(A \vee B) \vdash \exists x A, \exists x B \quad \exists_L$
6. $[c^1/x]A \vee [c^1/x]B \vdash \exists x A, \exists x B \quad \forall_L$
7. $[c^1/x]A \vdash \exists x A, \exists x B \quad \exists_R$
9. $[c^1/x]A \vdash [c^1/x]A, \exists x A, \exists x B \quad W_{R,R}$
10. $[c^1/x]A \vdash [c^1/x]A \quad (\text{акс.})$
8. $[c^1/x]B \vdash \exists x A, \exists x B \quad \exists_R$
11. $[c^1/x]B \vdash [c^1/x]B, \exists x A, \exists x B \quad W_{R,R}$
12. $[c^1/x]B \vdash [c^1/x]B \quad (\text{акс.})$

		(**)	
3.	$\vdash \exists x A \vee \exists x B \supset \exists x(A \vee B)$	$\supset_R$	
12.	$\exists x A \vee \exists x B \vdash \exists x(A \vee B)$	$\exists_R$	
13.	$\exists x A \vee \exists x B \vdash [t/x]A \vee [t/x]B, \exists x(A \vee B)$	W_R	
14.	$\exists x A \vee \exists x B \vdash [t/x]A \vee [t/x]B$	$\vee_R$	
15.	$\exists x A \vee \exists x B \vdash [t/x]A, [t/x]B$	$\vee_L$	
16.	$\exists x A \vdash [t/x]A, [t/x]B$	$\exists_L$	
18.	$\exists x A, [c_2/x]A \vdash [t/x]A, [t/x]B \quad (t = c_2)$		$\exists_L$
19.	$\exists x A, [c_2/x]A \vdash [c_2/x]A, [c_2/x]B \quad W_{R,L}$		
20.	$[c_2/x]A \vdash [c_2/x]A \quad (\text{акс.})$		
17.	$\exists x B \vdash [t/x]A, [t/x]B$		$\exists_L$
21.	$\exists x B, [c_2/x]B \vdash [t/x]A, [t/x]B \quad (t = c_2)$		
22.	$\exists x B, [c_2/x]B \vdash [c_2/x]A, [c_2/x]B \quad W_{R,L}$		
23.	$[c_2/x]B \vdash [c_2/x]B \quad (\text{акс.})$		

Задание 6 $\forall x A \wedge \exists x B \supset \exists x(A \wedge B)$

Доказательство.

1.	$\vdash \forall x A \wedge \exists x B \supset \exists x(A \wedge B)$	$\supset_R$	
2.	$\forall x A \wedge \exists x B \vdash \exists x(A \wedge B)$	$\wedge_L$	
3.	$\forall x A, \exists x B \vdash \exists x(A \wedge B)$	$\exists_L$	
4.	$\forall x A, [c_1/x]x B \vdash \exists x(A \wedge B)$	$\exists_R$	
5.	$\forall x A, [c_1/x]x B \vdash [c_1/x]A \wedge [c_1/x]B, \exists x(A \wedge B)$	$\forall_L$	
6.	$\forall x A, [c_1/x]A, [c_1/x]x B \vdash [c_1/x]A \wedge [c_1/x]B, \exists x(A \wedge B)$	$W_{L,R}$	
7.	$[c_1/x]A, [c_1/x]B \vdash [c_1/x]A \wedge [c_1/x]B$	$\wedge_R$	
8.	$[c_1/x]A, [c_1/x]B \vdash [c_1/x]A \quad W_L$		
10.	$[c_1/x]A \vdash [c_1/x]A \quad (\text{акс.})$		
9.	$[c_1/x]A, [c_1/x]B \vdash [c_1/x]B \quad W_L$		
11.	$[c_1/x]B \vdash [c_1/x]B \quad (\text{акс.})$		

Задание 7 $\forall x(A \supset B) \supset (\forall x(B \supset C) \supset \forall x(A \supset C))$

Сколемизация.

1.	$\forall x(A \supset B) \supset (\forall x(B \supset C) \supset \forall x(A \supset C))$	<i>Переименование x</i>
2.	$\forall x(A \supset B) \supset (\forall y(B \supset C) \supset \forall z(A \supset C))$	$a \supset b \equiv \neg a \vee b$
3.	$\exists x \neg(A \supset B) \vee (\exists y \neg(B \supset C) \vee \forall z(A \supset C))$	$a \supset b \equiv \neg a \vee b$
4.	$\exists x \neg(\neg A \vee B) \vee (\exists y \neg(\neg B \vee C) \vee \forall z(\neg A \vee C))$	
5.	$\exists x \neg(\neg A \vee B) \vee \exists y \forall z(\neg(\neg B \vee C) \vee (\neg A \vee C))$	
6.	$\exists x \exists y \forall z(\neg(\neg A \vee B) \vee \neg(\neg B \vee C) \vee (\neg A \vee C))$	<i>Де Морган</i>
7.	$\exists x \exists y \forall z(A \wedge \neg B \vee B \wedge \neg C \vee \neg A \vee C)$	$x \rightarrow c_1, y \rightarrow c_2$
8.	$\forall z(A \wedge \neg B \vee B \wedge \neg C \vee \neg A \vee C)$	
	<i>Закон $a \vee (b \wedge c) \equiv (a \vee b) \wedge (a \vee c)$</i>	
9.	$\forall z((A \vee \neg A) \wedge (\neg A \vee \neg B) \vee B \wedge \neg C \vee C)$	
	<i>Закон $a \vee (b \wedge c) \equiv (a \vee b) \wedge (a \vee c)$</i>	
10.	$\forall z((A \vee \neg A) \wedge (\neg A \vee \neg B) \vee (B \vee C) \wedge (\neg C \vee C))$	
	<i>Закон исключённого третьего</i>	
11.	$\forall z(\neg A \vee \neg B \vee B \vee C)$	
	<i>Получена сколемовская стандартная форма</i>	

Задание 8 $\exists x(F(x, y) \supset G(x, z)) \supset (\forall xG(x, z) \supset \exists xH(z, x))$
Сколемизация.

1. $\exists x(F(x, y) \supset G(x, z)) \supset (\forall xG(x, z) \supset \exists xH(z, x))$
Переименование каждого x
2. $\exists u(F(u, y) \supset G(u, z)) \supset (\forall vG(v, z) \supset \exists tH(z, t))$
Правило $a \supset b \equiv \neg a \vee b$
3. $\neg \exists u(\neg F(u, y) \vee G(u, z)) \vee (\neg \forall vG(v, z) \vee \exists tH(z, t))$
4. $\forall u \neg(\neg F(u, y) \vee G(u, z)) \vee (\exists v \neg G(v, z) \vee \exists tH(z, t))$
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5. $\forall u(F(u, y) \wedge \neg G(u, z)) \vee (\exists v \neg G(v, z) \vee \exists tH(z, t))$
6. $\forall u(F(u, y) \wedge \neg G(u, z)) \vee \exists v \exists t(\neg G(v, z) \vee H(z, t))$
7. $\forall u \exists v \exists t(F(u, y) \wedge \neg G(u, z) \vee \neg G(v, z) \vee H(z, t))$
8. $\forall y \forall z \forall u \exists v \exists t(F(u, y) \wedge \neg G(u, z) \vee \neg G(v, z) \vee H(z, t))$
 $v \rightarrow f(y, z, u), t \rightarrow g(y, z, u)$
9. $\forall y \forall z \forall u(F(u, y) \wedge \neg G(u, z) \vee \neg G(f(y, z, u), z) \vee H(z, g(y, z, u)))$
Закон $a \vee (b \wedge c) \equiv (a \vee b) \wedge (a \vee c)$
10. $\forall y \forall z \forall u((F(u, y) \vee \neg G(f(y, z, u), z)) \wedge (\neg G(u, z) \vee \neg G(f(y, z, u), z)) \vee H(z, g(y, z, u)))$
Закон $a \vee (b \wedge c) \equiv (a \vee b) \wedge (a \vee c)$
11. $\forall y \forall z \forall u((F(u, y) \vee \neg G(f(y, z, u), z) \vee H(z, g(y, z, u))) \wedge$
 $\wedge (\neg G(u, z) \vee \neg G(f(y, z, u), z) \vee H(z, g(y, z, u))))$
Получена сколемовская стандартная форма

Задание 9 *Задача: построить непротиворечивый набор из 3-х формул исчисления предикатов.*

Решение.

Рассмотрим набор формул: $\forall xP(x), \exists xQ(x), \exists x(P(x) \wedge Q(x))$

Рассмотрим интерпретацию:

$\mathbb{X} = \mathbb{N} = \{1, 2, 3, \dots\}$ — носитель алгебры
 $\mathbb{F} = \{\vee, \wedge, \supset\}$ — сигнатура алгебры
Получим алгебру: $\langle \mathbb{X}, \mathbb{F} \rangle$

$$P(x) := \frac{x}{x} = 1;$$

$$P(x) : \mathbb{X} \rightarrow \{0, 1\}$$

$$Q(x) := x \% 2 = 1;$$

$$Q(x) : \mathbb{X} \rightarrow \{0, 1\}$$

Формулы интерпретируются как:

- $\forall xP(x) = \forall x(\frac{x}{x} = 1)(1)$

Для любого x результат деления на x будет 1.

Истина.

- $\exists xQ(x) = \exists x(x \% 2 = 1)(2)$

Найдётся x , остаток от деления которого на 2 равен 1. Например: $3 \% 2 = 1$.

Истина.

- $\exists x(P(x) \wedge Q(x)) = \exists x(x \% 2 = 1)(3)$

Найдётся x , одновременно удовлетворяющий условиям (1) и (2). Например, $(\frac{3}{3} = 1) \wedge (3 \% 2 = 1)$.
Истина.

Таким образом, формулы истинны в построенной интерпретации $\implies$
 $\implies$ изначальный набор формул непротиворечив.

Задание 10 Задача: построить противоречивый набор из 3-х формул исчисления предикатов.

Решение.

Рассмотрим набор формул $\Gamma : \forall xA, \exists xB, \exists x(\neg A \wedge B)$

Проверим.

- $\Gamma \vdash \exists xA$

1. $\forall xA, \exists xB, \exists x(\neg A \wedge B) \vdash \exists xA$ $\exists_L, \wedge_L$
2. $\forall xA, \exists xB, \neg[c^1/x]A, [c^1/x]B \vdash \exists xA$ $\forall_L$
3. $\forall xA, [c^1/x]A, \exists xB, \neg[c^1/x]A, [c^1/x]B \vdash \exists xA$ $\exists_R$
4. $\forall xA, [c^1/x]A, \exists xB, \neg[c^1/x]A, [c^1/x]B \vdash [c^1/x]A, \exists xA$ $W_{L,L,L,R}$
5. $[c^1/x]A \vdash [c^1/x]A$ (акс.)

- $\Gamma \vdash \neg \exists xA$

1. $\forall xA, \exists xB, \exists x(\neg A \wedge B) \vdash \neg \exists xA$ $\exists_L, \wedge_L$
2. $\forall xA, \exists xB, \neg[c^1/x]A, [c^1/x]B \vdash \forall x \neg A$ $\forall_R$
3. $\forall xA, \exists xB, \neg[c^1/x]A, [c^1/x]B \vdash \forall x \neg A, \neg[c^1/x]A$ $W_{L,L,L,R}$
4. $\neg[c^1/x]A \vdash \neg[c^1/x]A$ (акс.)

Таким образом, было показано, что используя Γ можно доказать $\exists xA$ и $\neg \exists xA$. Следовательно, система Γ является противоречивой.